CBSE Class 12 Maths — Probability

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark (MCQ/Assertion-Reason/VSA)

Q1. Two numbers are selected at random from the integers 1 through 9. If the sum is even, find the probability that both numbers are odd. [PYQ 2018] [Confidence: High]

Q2. A die is rolled. If the outcome is an odd number, what is the probability that it is a prime number? [Practice] [Confidence: Medium]

Q3. If $P(A) = 0.4$, $P(B) = 0.8$, and $P(B|A) = 0.6$, find the value of $P(A \cap B)$. [PYQ 2020] [Confidence: Very High]

Q4. Let A and B be two independent events such that $P(A) = 0.3$ and $P(B) = 0.4$. Find the probability of the simultaneous occurrence of both events. [PYQ 2017] [Confidence: High]

Q5. A coin is tossed twice. If the outcome is at most one tail, what is the probability that both a head and a tail have appeared? [Practice] [Confidence: High]

Q6. The probability that person A hits a target is $\frac{1}{3}$ and the probability that person B hits it is $\frac{2}{5}$. What is the probability that the target will be hit if both person A and person B shoot at it? [Practice] [Confidence: High]

Q7. For two mutually exclusive events A and B, if $P(A) = \frac{1}{2}$ and $P(A \cup B) = \frac{3}{5}$, find $P(B)$. [PYQ 2019] [Confidence: Very High]

Q8. For two independent events A and B, if $P(A) = \frac{1}{2}$ and $P(A \cup B) = \frac{3}{5}$, find $P(B)$. [PYQ 2019] [Confidence: Very High]

Q9. An unbiased die is tossed twice. Find the probability of getting a $4, 5,$ or 6 on the first toss and a $1, 2, 3,$ or 4 on the second toss. [Practice] [Confidence: Medium]

Q10. What is the expected value (mean) of a binomial distribution with $n = 5$ trials and probability of success $p = \frac{1}{4}$? [PYQ 2022] [Confidence: High]

Q11. Write the sample space when a coin is tossed until a head appears. [Practice] [Confidence: Medium]

Q12. If $P(A) = 0.5$, $P(B) = 0$, what is the value of $P(A|B)$? [PYQ 2020] [Confidence: Very High]

Q13. If A and B are independent events with $P(A) = 0.3$ and $P(B) = 0.6$, find $P(\text{not } A \text{ and not } B)$. [PYQ 2018] [Confidence: High]

Q14. Find the variance of the number obtained on a single throw of an unbiased die. [PYQ 2017] [Confidence: High]

Q15. A loaded die has the following probabilities: $P(1) = P(2) = 0.2$, $P(3) = P(5) = P(6) = 0.1$, and $P(4) = 0.3$. Find the probability of getting an even number. [PYQ 2023] [Confidence: Very High]

Q16. Assertion (A): If A and B are independent events, then $P(A \cap B) = P(A)P(B)$. **Reason (R):** Mutually exclusive events are always independent. Choose the correct option. [PYQ 2024] [Confidence: Very High]

Q17. Assertion (A): The probability of an impossible event given that another event F has already occurred is 0. **Reason (R):** For any event F with $P(F) > 0$, $P(\emptyset|F) = 0$. Choose the correct option. [Practice] [Confidence: High]

Q18. Assertion (A): For any two events A and B such that $B \subset A$ and $P(B) \neq 0$, the conditional probability $P(A|B)$ is exactly 1. **Reason (R):** If $B \subset A$, then $A \cap B = B$. Choose the correct option. [Practice] [Confidence: High]

Q19. Assertion (A): The binomial distribution is applicable for the experiment of tossing a coin 3 times. **Reason (R):** Each toss results in exactly two possible outcomes and the probability of success remains constant across all tosses. Choose the correct option. [PYQ 2023] [Confidence: Very High]

Q20. If $P(A) = 0.4$, $P(B) = 0.6$, and $A \subset B$, find $P(A|B)$. [PYQ 2021] [Confidence: High]

2-Mark

Q21. A bag contains 5 red marbles and 3 black marbles. Three marbles are drawn one by one without replacement. What is the probability that at least one of the three marbles drawn is black, if the first marble drawn is known to be red? [PYQ 2019] [Confidence: Very High]

Q22. A die is thrown twice and the sum of the numbers appearing is observed to be 6. What is the conditional probability that the number 4 has appeared at least once? [PYQ 2020] [Confidence: Very High]

Q23. The probability of solving a specific problem independently by persons A and B are $\frac{1}{2}$ and $\frac{1}{3}$ respectively. If both try to solve the problem independently, find the probability that the problem is solved. [PYQ 2018] [Confidence: High]

Q24. An unbiased die is thrown twice. Let A be the event "odd number on the first throw" and B be the event "odd number on the second throw". Check the independence of the events A and B. [PYQ 2017] [Confidence: High]

Q25. The probabilities of two students A and B coming to the school in time are $\frac{3}{7}$ and $\frac{5}{7}$ respectively. Assuming that the events are independent, find the probability of only one of them coming to the school in time. [PYQ 2021] [Confidence: Medium]

Q26. Suppose you have two coins which appear identical in your pocket. You know that one is fair and one is 2-headed. If you take one out, toss it and get a head, what is the probability that it was a fair coin? [PYQ 2022] [Confidence: Very High]

Q27. Find the probability of throwing at most 2 sixes in 6 throws of a single unbiased die. (Write the expression using combinatorial notation). [Practice] [Confidence: Medium]

Q28. A random variable X takes the values $0, 1, 2$ with probabilities $\frac{3}{10}, \frac{2}{5}, \frac{3}{10}$ respectively. Calculate the expected value $E(X)$. [PYQ 2016] [Confidence: High]

Q29. If A and B are two independent events, mathematically prove that their complements A' and B' are also independent events. [PYQ 2019] [Confidence: High]

Q30. Let $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$. Find the value of $P(A|B')$. [PYQ 2023] [Confidence: Very High]

Q31. In a dice game, a player pays a stake of ₹1 for each throw of a die. She receives ₹5 if the die shows a 3, ₹2 if the die shows a 1 or 6, and nothing otherwise. What is the player's expected profit per throw over a long series of throws? [PYQ 2020] [Confidence: High]

Q32. Let A and B be independent events such that $P(A) = 0.3$ and $P(A \cup B) = 0.5$. Find $P(B)$. [Practice] [Confidence: Medium]

3-Mark

Q33. Five cards are drawn successively from a well-shuffled pack of 52 cards with replacement. Determine the probability that (i) all the five cards are spades, (ii) only 3 cards are spades. [PYQ 2018] [Confidence: Very High]

Q34. In a hurdle race, a player has to cross 10 hurdles. The probability that he will clear each hurdle is $\frac{5}{6}$. What is the probability that he will knock down fewer than 2 hurdles? [PYQ 2022] [Confidence: Very High]

Q35. A die is loaded in such a way that an even number is twice as likely to occur as an odd number. If the die is tossed twice, find the probability distribution of the random variable representing the number of perfect squares in the two tosses. [PYQ 2019] [Confidence: High]

Q36. A man is known to speak the truth 8 times out of 10 times. A die is tossed. He reports that it was a 5. What is the probability that it was actually a 5? [PYQ 2017] [Confidence: Very High]

Q37. A letter is known to have come either from TATA NAGAR or from CALCUTTA. On the envelope, just two consecutive letters 'TA' are visible. What is the exact probability that the letter came from TATA NAGAR? [PYQ 2021] [Confidence: High]

Q38. A bag contains $(2n + 1)$ coins. It is known that n of these coins have a head on both sides whereas the rest of the coins are fair. A coin is picked up at random from the bag and is tossed. If the probability that the toss results in a head is $\frac{31}{42}$, determine the value of n. [Practice] [Confidence: Medium]

Q39. On a multiple choice examination with 3 possible answers (out of which only one is correct) for each of the 5 questions, what is the probability that a candidate would get 4 or more correct answers just by guessing? [PYQ 2020] [Confidence: High]

Q40. How many times must a man toss a fair coin so that the probability of having at least one head is strictly more than 80%? [PYQ 2018] [Confidence: High]

Q41. Let X denote the number of hours you study on a Sunday. It is known that $P(X = x) = kx$ for $x = 1, 2$, $P(X = x) = k(5 - x)$ for $x = 3, 4$, and 0 otherwise, where k is a constant. Find the value of k and the probability that you study at least two hours. [PYQ 2023] [Confidence: Very High]

Q42. Two persons A and B throw a die alternatively till one of them gets a 6 and wins the game. Find their respective probabilities of winning if A starts the game first. [PYQ 2019] [Confidence: High]

5-Mark (Long Answer/Case-Study)

Q43. In a factory, which manufactures nuts, machines A, B, and C manufacture respectively 25%, 35%, and 40% of the nuts. Of their output, 5%, 4%, and 2% respectively are defective nuts. A nut is drawn at random from the product and is found to be defective. Find the probability that it is manufactured by machine B. [PYQ 2020] [Confidence: Very High]

Q44. In answering a question on a multiple-choice test with 4 choices per question, a student knows the answer, guesses, or copies the answer. Let $\frac{1}{2}$ be the probability that he knows the answer, $\frac{1}{4}$ be the probability that he guesses, and $\frac{1}{4}$ that he copied it. Assuming that a student who copies the answer will be correct with a probability of $\frac{3}{4}$, and a student who guesses is correct with probability $\frac{1}{4}$, what is the exact probability that the student knew the answer, given that he answered it correctly? [PYQ 2018] [Confidence: Very High]

Q45. An insurance company insured $2,000$ cyclists, $4,000$ scooter drivers, and $6,000$ motorbike drivers. The probability of an accident involving a cyclist, a scooter driver, and a motorbike driver are 0.01, 0.03, and 0.15 respectively. One of the insured persons meets with an accident. What is the probability that he is a scooter driver? [PYQ 2019] [Confidence: Very High]

Q46. Three machines E_1, E_2, and E_3 in a certain factory produce 50%, 25%, and 25%, respectively, of the total daily output of electric tubes. It is known that 4% of the tubes produced on machines E_1 and E_2 are defective, and that 5% of those produced on E_3 are defective. If one tube is picked up at random from a day's production, calculate the probability that it is defective. Furthermore, if the selected tube is found to be defective, what is the probability it came from machine E_1? [PYQ 2022] [Confidence: High]

Q47. Case Study: A company conducts a mandatory health checkup for all newly hired employees to check for a blood infection. The blood infection affects roughly 5% of the population. The probability of a false positive on the test for this infection is 4% (meaning $P(\text{Positive}|\text{Not Infected}) = 0.04$), while the probability of a false negative on the test is 3% (meaning $P(\text{Negative}|\text{Infected}) = 0.03$). (i) Find the total probability that a randomly chosen new employee tests positive. (2 Marks) (ii) If a person tests positive for the infection, use Bayes' theorem to calculate the probability that they are actually infected. (3 Marks) [Practice] [Confidence: Very High]

Q48. Case Study: A survey says that approximately 30% of all products ordered online across various e-commerce websites are returned. Two products that are ordered online are selected at random. Let X represent the number of products that are returned out of the two. (i) Find the probability distribution of the random variable X. (3 Marks) (ii) Calculate the mean (expectation) of X. (2 Marks) [Practice] [Confidence: High]

Q49. Case Study: Bag I contains 3 red and 4 black balls. Bag II contains 5 red and 6 black balls. One bag is chosen at random and a ball is drawn from it. (i) Find the total probability that the drawn ball is red. (2 Marks) (ii) If the drawn ball is observed to be red, find the exact probability that it was drawn from Bag I. (3 Marks) [PYQ 2021] [Confidence: Very High]

Q50. Case Study: Out of a group of 8 highly qualified doctors in a hospital, 6 are very kind and cooperative with their patients and so are very popular, while the other 2 remain reserved. For a health camp, 3 doctors are selected at random from this group. (i) Find the probability distribution of the number of very popular doctors selected for the camp. (3 Marks) (ii) What is the expected number of popular doctors in the selected committee? (2 Marks) [PYQ 2017] [Confidence: Medium]

Answer Key

Q1. $S = \{(1,3),(1,5),(1,7),(1,9),(2,4),(2,6),(2,8),\dots\}$. Number of ways to pick 2 numbers so sum is even = $5C_2$ (both odd) + $4C_2$ (both even) = $10+6=16$. Number of ways both are odd = 10. Probability = $\frac{10}{16}=\frac{5}{8}$. **Q2.** Odd numbers on a die: $\{1,3,5\}$. Prime odd numbers: $\{3,5\}$. Probability = $\frac{2}{3}$. **Q3.** $P(A\cap B)=P(A)\times P(B|A)=0.4\times 0.6=0.24$. **Q4.** For independent events, $P(A\cap B)=P(A)\times P(B)=0.3\times 0.4=0.12$. **Q5.** Sample space for at most one tail = $\{HH,HT,TH\}$. Favorable (both head and tail) = $\{HT,TH\}$. Probability = $\frac{2}{3}$. **Q6.** $P(\text{Target Hit})=1-P(\text{None Hit})=1-P(A')P(B')=1-(\frac{2}{3}\times\frac{3}{5})=1-\frac{2}{5}=\frac{3}{5}$. **Q7.** For mutually exclusive events, $P(A\cup B)=P(A)+P(B)\Rightarrow\frac{3}{5}=\frac{1}{2}+P(B)\Rightarrow P(B)=\frac{1}{10}$. **Q8 दल.** For independent events, $P(A\cup B)=P(A)+P(B)-P(A)P(B)\Rightarrow\frac{3}{5}=\frac{1}{2}+P(B)-\frac{1}{2}P(B)\Rightarrow\frac{1}{10}=\frac{1}{2}P(B)\Rightarrow P(B)=\frac{1}{5}$. **Q9.** $P(\text{first is 4, 5, or 6})=\frac{3}{6}=\frac{1}{2}$. $P(\text{second is 1, 2, 3, or 4})=\frac{4}{6}=\frac{2}{3}$. Probability = $\frac{1}{2}\times\frac{2}{3}=\frac{1}{3}$. **Q10.** Mean $E(X)=np=5\times\frac{1}{4}=\frac{5}{4}=1.25$. **Q11.** $S=\{H,TH,TTH,TTTH,\dots\}$. **Q12.** $P(A|B)=\frac{P(A\cap B)}{P(B)}$. Since $P(B)=0$, $P(A|B)$ is not defined. **Q13.** $P(A'\cap B')=P(A')P(B')=(1-0.3)(1-0.6)=0.7\times 0.4=0.28$. **Q14.** $E(X)=3.5$. $E(X^2)=\frac{1^2+2^2+3^2+4^2+5^2+6^2}{6}=\frac{91}{6}$. Variance = $E(X^2)-[E(X)]^2=\frac{91}{6}-\frac{49}{4}=\frac{35}{12}$. **Q15.** $P(\text{even})=P(2)+P(4)+P(6)=0.2+0.3+0.1=0.6$. **Q16.** Option C (A is true, but R is false. Mutually exclusive events are highly dependent since if one happens, the other cannot). **Q17.** Option A (Both A and R are true, and R is the correct explanation of A). **Q18.** Option A (Both A and R are true, and R is the correct explanation of A. $P(A|B)=\frac{P(A\cap B)}{P(B)}=\frac{P(B)}{P(B)}=1$). **Q19.** Option A (Both A and R are true, and R is the correct explanation of A). **Q20.** $P(A|B)=\frac{P(A\cap B)}{P(B)}$. Since $A\subset B$, $A\cap B=A$. So, $P(A|B)=\frac{P(A)}{P(B)}=\frac{0.4}{0.6}=\frac{2}{3}$.

Q21. After drawing 1 red, 4 red and 3 black remain (7 total). Two more are drawn. $P(\text{no black in next two})=\frac{\binom{4}{2}}{\binom{7}{2}}=\frac{6}{21}=\frac{2}{7}$. $P(\text{at least one black})=1-\frac{2}{7}=\frac{5}{7}$. **Q22.** Sample space for sum 6: $S'=\{(1,5),(5,1),(2,4),(4,2),(3,3)\}$ (5 outcomes). Outcomes with a 4: $E'=\{(2,4),(4,2)\}$ (2 outcomes). $P=\frac{2}{5}$. **Q23.** $P(\text{Problem Solved})=1-P(\text{None solves})=1-(1-\frac{1}{2})(1-\frac{1}{3})=1-(\frac{1}{2}\times\frac{2}{3})=1-\frac{1}{3}=\frac{2}{3}$. **Q24.** $P(A)=\frac{18}{36}=\frac{1}{2}$, $P(B)=\frac{18}{36}=\frac{1}{2}$. $P(A\cap B)=P(\text{odd on both})=\frac{9}{36}=\frac{1}{4}$. Since $P(A\cap B)=P(A)P(B)$, events A and B are independent. **Q25.** $P(\text{only one})=P(A)P(B')+P(A')P(B)=(\frac{3}{7}\times\frac{2}{7})+(\frac{4}{7}\times\frac{5}{7})=\frac{6}{49}+\frac{20}{49}=\frac{26}{49}$. **Q26.** $P(H|F)=\frac{1}{2}$, $P(H|T)=1$. Using Bayes' Theorem: $P(F|H)=\frac{P(F)P(H|F)}{P(F)P(H|F)+P(T)P(H|T)}=\frac{0.5\times0.5}{0.5\times0.5+0.5\times1}=\frac{0.25}{0.75}=\frac{1}{3}$. **Q27.** Using Binomial Distribution $B(6,\frac{1}{6})$: $P(X\le 2)=\binom{6}{0}(\frac{5}{6})^6+\binom{6}{1}(\frac{1}{6})(\frac{5}{6})^5+\binom{6}{2}(\frac{1}{6})^2(\frac{5}{6})^4$. **Q28.** $E(X)=\sum x_ip_i=0(\frac{3}{10})+1(\frac{2}{5})+2(\frac{3}{10})=0+\frac{4}{10}+\frac{6}{10}=\frac{10}{10}=1$. **Q29.** $P(A'\cap B')=P((A\cup B)')=1-P(A\cup B)=1-[P(A)+P(B)-P(A\cap B)]$. Given $P(A\cap B)=P(A)P(B)$, it becomes $1-P(A)-P(B)+P(A)P(B)=(1-P(A))(1-P(B))=P(A')P(B')$. Thus independent. **Q30.** $P(A|B')=\frac{P(A\cap B')}{P(B')}=\frac{P(A)-P(A\cap B)}{1-P(B)}=\frac{0.6-0.2}{1-0.5}=\frac{0.4}{0.5}=\frac{4}{5}$. **Q31.** X is profit. If 3 rolls, win ₹5, profit is ₹4. If 1 or 6 rolls, win ₹2, profit is ₹1. Else win ₹0, profit is -₹1. Expected profit Math input error. **Q32.** $P(A\cup B)=P(A)+P(B)-P(A)P(B)\Rightarrow 0.5=0.3+P(B)-0.3P(B)\Rightarrow 0.2=0.7P(B)\Rightarrow P(B)=\frac{2}{7}$.

Q33. Let $p=\frac{1}{4}$ (spade), $q=\frac{3}{4}$. (i) All spades = $\binom{5}{5}(\frac{1}{4})^5=\frac{1}{1024}$. (ii) Only 3 spades = $\binom{5}{3}(\frac{1}{4})^3(\frac{3}{4})^2=10\times\frac{1}{64}\times\frac{9}{16}=\frac{45}{512}$. **Q34.** Let $p=\frac{5}{6}$ (clear), $q=\frac{1}{6}$ (knock down). Knock down random variable $Y\sim B(10,\frac{1}{6})$. $P(Y<2)=P(Y=0)+P(Y=1)=\binom{10}{0}(\frac{5}{6})^{10}+\binom{10}{1}(\frac{1}{6})^1(\frac{5}{6})^9=(\frac{5}{6})^9(\frac{5}{6}+\frac{10}{6})=\frac{15}{6}(\frac{5}{6})^9=\frac{5}{2}(\frac{5}{6})^9$. **Q35.** $P(\text{Even})=\frac{2}{3}$, $P(\text{Odd})=\frac{1}{3}$. Perfect squares are 1 and 4. $P(1)=P(\text{Odd and Square})=\frac{1}{3}$ of $P(\text{Odd})=\frac{1}{9}$. $P(4)=P(\text{Even and Square})=\frac{1}{3}$ of $P(\text{Even})=\frac{2}{9}$. Thus $P(\text{Square})=\frac{1}{9}+\frac{2}{9}=\frac{1}{3}$.

$X \sim B(2, \frac{1}{3})$. $P(0) = \frac{4}{9}, P(1) = \frac{4}{9}, P(2) = \frac{1}{9}$. **Q36.** $P(T) = \frac{4}{5}, P(F) = \frac{1}{5}$. $P(\text{Report 5}|\text{Is 5}) = \frac{4}{5}$, $P(\text{Report 5}|\text{Not 5}) = \frac{1}{5}$. $P(\text{Is 5}) = \frac{1}{6}$, $P(\text{Not 5}) = \frac{5}{6}$. Bayes: $\frac{\frac{1}{6}\times\frac{4}{5}}{\frac{1}{6}\times\frac{4}{5}+\frac{5}{6}\times\frac{1}{5}} = \frac{4}{4+5} = \frac{4}{9}$. **Q37.** TATA NAGAR has 9 letters, 8 consecutive pairs, 2 are "TA" $\Rightarrow P(TA|T) = \frac{2}{8} = \frac{1}{4}$. CALCUTTA has 8 letters, 7 pairs, 1 is "TA" $\Rightarrow P(TA|C) = \frac{1}{7}$. Bayes: $P(T|TA) = \frac{0.5\times0.25}{0.5\times0.25+0.5\times(1/7)} = \frac{\frac{1}{4}}{\frac{1}{4}+\frac{1}{7}} = \frac{7}{11}$. **Q38.** $P(H) = \frac{n}{2n+1}(1) + \frac{n+1}{2n+1}(\frac{1}{2}) = \frac{3n+1}{4n+2}$. Equating: $\frac{3n+1}{4n+2} = \frac{31}{42} \Rightarrow 126n + 42 = 124n + 62 \Rightarrow 2n = 20 \Rightarrow n = 10$. **Q39.** $B(5, \frac{1}{3})$. $P(X \geq 4) = \binom{5}{4}(\frac{1}{3})^4(\frac{2}{3})^1 + \binom{5}{5}(\frac{1}{3})^5 = \frac{10}{243} + \frac{1}{243} = \frac{11}{243}$. **Q40.** $1 - (\frac{1}{2})^n > 0.8 \Rightarrow 0.2 > (\frac{1}{2})^n \Rightarrow 2^n > 5 \Rightarrow n \geq 3$. He must toss at least 3 times. **Q41.** Sum of probabilities = $1 \Rightarrow k + 2k + 2k + k = 1 \Rightarrow 6k = 1 \Rightarrow k = \frac{1}{6}$. $P(X \geq 2) = P(2) + P(3) + P(4) = 5k = \frac{5}{6}$. **Q42.** $P(\text{A wins}) = \frac{1}{6} + (\frac{5}{6})^2\frac{1}{6} + (\frac{5}{6})^4\frac{1}{6} + \cdots = \frac{1/6}{1-25/36} = \frac{1/6}{11/36} = \frac{6}{11}$. $P(\text{B wins}) = 1 - \frac{6}{11} = \frac{5}{11}$.

Q43. $P(A) = 0.25, P(B) = 0.35, P(C) = 0.40$. Defect rates: $P(D|A) = 0.05, P(D|B) = 0.04, P(D|C) = 0.02$. $P(B|D) = \frac{0.35\times0.04}{0.25\times0.05+0.35\times0.04+0.40\times0.02} = \frac{0.014}{0.0125+0.014+0.008} = \frac{0.014}{0.0345} = \frac{140}{345} = \frac{28}{69}$. **Q44.** $P(Know) = 0.5, P(Guess) = 0.25, P(Copy) = 0.25$. $P(Correct|Know) = 1, P(Correct|Guess) = 0.25, P(Correct|Copy) = 0.75$. $P(Know|Correct) = \frac{0.5\times1}{0.5\times1+0.25\times0.25+0.25\times0.75} = \frac{0.5}{0.5+0.0625+0.1875} = \frac{0.5}{0.75} = \frac{2}{3}$. **Q45.** Total drivers $= 12,000$. $P(C) = \frac{2}{12} = \frac{1}{6}, P(S) = \frac{4}{12} = \frac{1}{3}, P(M) = \frac{6}{12} = \frac{1}{2}$. $P(A|C) = 0.01, P(A|S) = 0.03, P(A|M) = 0.15$. $P(S|A) = \frac{\frac{1}{3}\times0.03}{\frac{1}{6}\times0.01+\frac{1}{3}\times0.03+\frac{1}{2}\times0.15} = \frac{0.01}{0.00167+0.01+0.075} = \frac{0.01}{0.08667} = \frac{120}{1040} = \frac{3}{26}$. **Q46.** (i) $P(Defective) = (0.50 \times 0.04) + (0.25 \times 0.04) + (0.25 \times 0.05) = 0.020 + 0.010 + 0.0125 = 0.0425$ (or 4.25%). (ii) $P(E_1|Defective) = \frac{0.020}{0.0425} = \frac{200}{425} = \frac{8}{17}$. **Q47.** (i) $P(\text{Infected}) = 0.05$, $P(\text{Not}) = 0.95$. $P(\text{Pos}|\text{Inf}) = 1 - 0.03 = 0.97$. $P(\text{Pos}|\text{Not}) = 0.04$. Total $P(\text{Pos}) = (0.05 \times 0.97) + (0.95 \times 0.04) = 0.0485 + 0.0380 = 0.0865$. (ii) $P(\text{Inf}|\text{Pos}) = \frac{0.0485}{0.0865} = \frac{485}{865} = \frac{97}{173}$. **Q48.** (i) $X \sim B(2, 0.3)$. $P(X = 0) = (0.7)^2 = 0.49$. $P(X = 1) = 2(0.3)(0.7) = 0.42$. $P(X = 2) = (0.3)^2 = 0.09$. (ii) Mean $E(X) = \sum x_i p_i = 0(0.49) + 1(0.42) + 2(0.09) = 0.42 + 0.18 = 0.60$ (or $E(X) = np = 2 \times 0.3 = 0.6$). **Q49.** (i) $P(\text{Red}) = P(\text{Bag I})P(\text{Red}|\text{I}) + P(\text{Bag II})P(\text{Red}|\text{II}) = \frac{1}{2}(\frac{3}{7}) + \frac{1}{2}(\frac{5}{11}) = \frac{3}{14} + \frac{5}{22} = \frac{33+35}{154} = \frac{68}{154} = \frac{34}{77}$. (ii) $P(\text{Bag I}|\text{Red}) = \frac{3/14}{34/77} = \frac{3}{14} \times \frac{77}{34} = \frac{33}{68}$. **Q50.** (i) Let X be number of popular doctors. Total docs = 8. 3 selected. Reserved = 2. So X can be $1, 2, 3$. $P(X = 1) = \frac{\binom{6}{1}\binom{2}{2}}{\binom{8}{3}} = \frac{6}{56} = \frac{3}{28}$. $P(X = 2) = \frac{\binom{6}{2}\binom{2}{1}}{\binom{8}{3}} = \frac{30}{56} = \frac{15}{28}$. $P(X = 3) = \frac{\binom{6}{3}\binom{2}{0}}{\binom{8}{3}} = \frac{20}{56} = \frac{10}{28}$. (ii) Mean $= 1(\frac{3}{28}) + 2(\frac{15}{28}) + 3(\frac{10}{28}) = \frac{3+30+30}{28} = \frac{63}{28} = \frac{9}{4} = 2.25$.
